HHAI 2025
D. Pedreschi et al. (Eds.)
© 2025 The Authors.
This article is published online with Open Access by IOS Press and distributed under the terms of the Creative Commons Attribution Non-Commercial License 4.0 (CC BY-NC 4.0).
doi:10.3233/FAIA250626

Hybrid Human-AI Systems for Preserving Cultural Heritage and Craftsmanship

Alexandra MCDOUGALL[a,] and Hortense GERARDO[b1]
[a] *SmartBear Software*
[b] *University of California, San Diego*
ORCiD ID: Alexandra McDougall https://orcid.org/0009-0006-4854-2353
ORCiD ID: Hortense Gerardo https://orcid.org/0000-0002-3436-0763

Abstract. The rapid advancement of Artificial Intelligence (AI) presents unique opportunities for preserving and revitalizing cultural heritage and traditional craftsmanship. This paper proposes a novel approach to training AI systems using archival documentary footage, specifically focusing on ensuring ethical data practices and empowering Indigenous communities. Our approach is grounded in the Global Indigenous Data Alliance's "Indigenous Peoples' Rights in Data" framework, which outlines 12 key principles for ethical data governance.[1] These principles, including "the rights to possess, consent, refuse, benefit, and use data originating from a given community," inform our proposed methodology. By providing clear guidelines on data governance, we aim to encourage traditionally closed cultures to share their practices, ensuring that the preservation efforts benefit the communities themselves. AI systems trained on archival footage and informed by ethical data practices can be seen as a state-of-the-art form of salvage ethnography and empower communities to maintain control over their traditional knowledge and cultural heritage. This approach has the potential to ensure that diverse artistic practices and traditional craftsmanship continue to thrive in the age of AI.

Keywords. Artificial Intelligence, craftsmanship, salvage ethnography, robotics

1. Introduction

"Every sensation of every being of the world is a mode through which the world lives and feels itself," writes Vinciane Despret. "When a being is no more, the world narrows all of a sudden, and a part of reality collapses."[2] This poignant observation underscores the urgency of our mission to preserve cultural heritage and traditional craftsmanship in the face of rapid technological change and cultural erosion.

A prime example of a human-AI partnership in cultural preservation is the SkoBot project, developed by Danielle Boyer, an Ojibwe innovator.[3] This project utilizes AI models trained on language recordings from community Elders and experts to create interactive, wearable robots that help preserve indigenous languages and cultural practices. This approach demonstrates how AI can be a powerful tool for cultural continuity when developed in collaboration with and for the benefit of indigenous communities.

[1] Corresponding Author: Hortense Gerardo, hgerardo@ucsd.edu.

Various data sources are crucial in developing diverse and culturally sensitive AI systems. AI companies are acquiring video footage from content creators to train their tools.[4] We propose extending this practice to include archival documentary footage, focusing on materials that document traditional craftsmanship and cultural practices.

Recent developments in AI-powered visual recognition have allowed companies to dramatically improve searchability within video content libraries.[5] While this allows for bringing endangered cultural traditions into the AI age with relative technological ease, we must put guardrails in place to avoid alienating the peoples from whom the traditions originated.

Data governance is the cornerstone of collaborating with those most at risk—Indigenous cultures and tradespeople—regarding monetizable skills that robotics companies will seek to emulate. Some skills—such as sewing—are so broadly practiced that no group claim them[6], and thus, data governance will do little to protect those whose livelihoods will be threatened.

Crafts created by specialists without the next generation to take up the tradition are at risk of disappearance: this runs the gamut from Arizapana, the last in a 500-year lineage of Incan bridge weavers[7], to crafting parchment and vellum.[8]

While thorough documentation can help researchers recreate lost traditions, such as Tim Severin's "The Brendan Voyage," which depicts him building a leather curragh to take him across the Atlantic[9], for complex processes, demonstration tends to be better than documentation because visual illustrations of complex concepts allow for easier understanding by breaking processes into manageable steps.

We propose an on-premise knowledge base owned by the craftspeople or the culture from which the craft and any subsequent training materials originated. The knowledge base will be returned to humanity as a whole upon the retirement of the last trade practitioner or at a time of the originating culture's choosing. We seek to eliminate data harm, "the adverse effects caused by uses of data that damage or set back a person, group, entity or society's interests."[10] For a comprehensive description of data harm, see Redden, Brand & Terzieva[11].

2. Methodology for AI Training on Archival Footage

2.1. Data Collection

Identify and acquire documentary videos designed to preserve or teach craftsmanship, such as historical reenactment series. An excellent example of this is the historical documentary TV series Victorian Farm.[12] In the series, historian Ruth Goodman and archaeologists Peter Ginn and Alex Langlands move into a Victorian smallholding to live the life of Victorian farmers, wearing period clothes and only using materials that would have been available in 1885. The team recreated historical recipes, processes, and crafts by referencing period sources. Topics covered in the series include stain removal, threshing wheat, plowing fields, cider-making, pest control, cheese-making, straw plaiting, and brewing beer.

2.2. Data Processing

Utilize digital tools, such as Adobe Premiere's Media Intelligence or Linius' Whizzard Portal[2], to name a few of the many such available resources, to scan and segment videos, tagging them with relevant metadata for efficient retrieval and analysis.

2.3. Depth Perception Enhancement

Process video clips through Scale-Invariant Monocular Depth Estimation tools to provide AI systems with crucial depth perception information, enhancing their understanding of physical processes. "Scale-Invariant Monocular Depth Estimation via SSI Depth" introduces a novel approach to estimating depth from a single image.[13][14] This method allows a computer to understand the 3D structure of a scene from just one 2D picture without knowing the actual size of objects in the image. The key innovation of this approach is its ability to produce accurate depth maps without requiring information about the scale of objects in the scene.[14] This innovation is beneficial because it allows the system to work with a wide variety of images, even when the size of objects is unknown or varies significantly. While depth perception is valuable for object manipulation[15][16], environmental understanding, transfer learning, and data augmentation[17], it is not the only factor in training robots from video content. DeepMind's research demonstrates that other approaches, such as reinforcement learning and learning from demonstrations, can also be practical in developing dexterous robotic behaviors.[18] These methods can complement depth perception to create more robust and capable robotic systems.

2.4. Ethical Data Governance

Implement strict protocols based on the Indigenous Peoples' Rights in Data framework to ensure community ownership and control over cultural data. The rights outlined by the Global Indigenous Data Alliance fall under two categories: 1) 'Data for Governance,' which primarily relates to the ability of Indigenous communities to access and use data themselves, and 2) 'Governance of Data,' which relates to the ability of Indigenous communities to steward and externally influence the use of data internally.[19]

3. Potential Application and Benefits

3.1. Ethical Data Governance

AI systems trained on archival footage can help document and recreate traditional crafting techniques at risk of being lost. Some uncommon crafts may not seem immediately practical but may be the foundation upon which future cultural ratcheting can be based. One example is the South Pacific practice of spiderweb fishing, where a spider web is used as a fishing line. Often, the webs are attached to kites to create a "lure" that skims across the water's surface.[20]

[2] In the interest of full disclosure, one of the authors is a minor investor in Linius.

3.2. Sustainable Production

AI-assisted crafting could promote more sustainable production methods, reducing reliance on mass-produced goods and their associated environmental impacts. A prime example of this is pasta; if a robot kitchen can form various pasta shapes from scratch, then people with access to the robot kitchen will not need to purchase mass-produced boxes of dried pasta.

3.3. Cultural Education

Interactive AI systems could serve as educational tools, providing immersive experiences in traditional craftsmanship and cultural practices. An example of this concept put into practice for a more modern application is the use of humanoid Aura robots installed in the MSG Sphere in Las Vegas.[21] There is an opportunity to use humanoid robots as ambassadors of ethnographic salvage for celibate communities like the Shakers.[22] Only two practicing Shakers are left today, Brother Arnold Hadd and Sister June Carpenter; the younger of the two (Hadd) is 67.[23]

3.4. Health and Safety

By capturing the expertise of master craftspeople, AI could help mitigate the physical risks associated with specific crafts, preserving the knowledge while protecting practitioners from potential health hazards, such as silicosis in relation to stonemasonry[24], industrial hazards such as carcinogenic exposure to hexavalent chromium[25], and arthritis.[26]

3.5. Co-Invention

In Melissa A. Schilling's book Quirky: The Remarkable Story of the Traits, Foibles, and Genius of Breakthrough Innovators Who Changed the World, she codifies the characteristics that tend to lead individual people to make extraordinary breakthroughs.[27] While some of the characteristics are not specifically relevant to a machine, for example, "opportunities and challenges of an era," AI mimics essential attributes of human inventors. Hallucinations in AI are interpreted as unfavorable, but they can also be analyzed through the lens of a creative breakthrough.[28] The sense of "separateness" and extreme confidence described by Ms. Schilling are hallmarks of LLMs. Robots trained on a vast array of artifacts of human culture have a broader corpus of knowledge from which to draw. The collaboration of innovative humans with a robotic assistant trained on a broad set of data points yields a connection far more than the sum of its parts.

4. Potential Unintended Effects

4.1. Individual Autonomy and Withholding

One possible complication is the decision by a "last practitioner" to withhold knowledge, even when predecessors may have consented to share. This choice should be

acknowledged as a legitimate exercise of individual autonomy and may reflect complex intergenerational, political, or spiritual concerns. From an ethical standpoint, our methodology must not override such decisions. Doing so would not only violate the principle of consent but could also fracture community trust and disrupt nuanced traditions of knowledge stewardship.

The right to refuse—outlined in the Global Indigenous Data Alliance framework—is not only collective but personal. AI systems must be designed with mechanisms that accommodate such refusals, including the possibility that no data may be available or appropriate for training. This, too, is a form of cultural preservation: one that honors the sovereignty of silence and restraint.

4.2. Risks of Cultural Essentialism and Fixity

There is also a risk that codifying traditional knowledge into an AI system could inadvertently fossilize practices that are, by nature, evolving and dynamic. Craftsmanship often involves improvisation, personal flair, and adaptation over time. When AI systems learn from archival footage or static documentation, they may reduce these practices to a fixed "style" or "form," losing the living dimension of the craft. This essentialism may be particularly problematic if the AI-generated representations are later used without community oversight or recontextualization, leading to the creation of "authenticity standards" that were never meant to exist.

4.3. Data Misuse and Misappropriation

Despite the best efforts to follow ethical data governance principles, the possibility remains that once encoded, cultural knowledge could be used beyond its intended scope. For example, generative AI models may synthesize culturally significant artifacts without appropriate attribution or permissions. These outputs could then be commercialized or widely disseminated, leading to a new form of digital misappropriation. To mitigate this, ongoing oversight and enforceable safeguards must be developed in collaboration with the source communities. Ethical preservation is not a one-time agreement, but a sustained partnership grounded in trust, reciprocity, and respect for context.

5. Challenges and Considerations

While this approach offers significant potential for cultural preservation and AI advancement, it also presents challenges.

5.1. Balancing Tradition and Technology

Efforts must be made to ensure that AI assistance preserves rather than replaces traditional practices. When advancement rapidly changes processes, replacing manual labor with faster technology, income sources for the displaced workers may dry up. Historically, this has contributed to riots and social upheavals. "One implication is that redistribution matters…policymakers must consider how to support the people who are most affected. Otherwise, political and social unrest becomes more likely, perhaps

inevitable."[29] If "The underlying purpose of AI is to allow wealth to access skill while removing from the skilled the ability to access wealth,"[30] how do we as a society ensure that skilled workers continue to make a living wage?

5.2. Data Sovereignty

Informed and culturally sensitive adherence to ethical data practices is essential to earning trust and respect for cultural identity. It includes respecting cultural bearers' inclinations to withhold secrets of their craft, as exemplified by Chiara Vigo's stance against monetizing Byssus.[31] One way to address this might be to introduce AI literacy within Indigenous communities. The introduction of AI literacy programs that enable an understanding of its capabilities, limitations, and ethical risks, may encourage community members to recognize the proposed system as a tool of empowerment rather than a mechanism for digital colonialism.

5.3. Technological Limitations

Current AI systems may struggle to fully capture traditional craftsmanship's nuances and tacit knowledge. For example, machines cannot properly replicate specific pasta shapes. Trofie is a short, twisted pasta thick in the middle, with pointed ends. While it has pointed ends, machine-produced trofie has little difference in the width over the shape of each piece. Using the pad of one's palm allows hand-made trofie to be shaped into a geometry resembling a conch shell.[32] Geometric differences are known to affect everything from cook times to sauce pairings. Since the pressure that a hand exerts against pasta dough being pressed into a cutting board is not easy to measure in video form, it is likely that our proposal will not capture some nuances.

5.4. Ethical Dilemmas

In cases where the last practitioner of a craft decides to invoke their right to refuse to share knowledge, that decision will inevitably be challenged. We must be prepared to honor such choices, even if it means losing connection to ancient traditions. For example, Chiara Vigo is the last person who makes Byssus (sea silk) from the spit of the endangered Pinna nobilis clam.[31] She is accompanied by the Italian coastguard in order for her to dive and harvest the thin fibers growing from the tips of the clams. While she has trained at least one of her grandchildren in the art of how to craft Byssus, her daughter Maddalena has seemingly chosen a different path in life. Mrs. Vigo's nine-year-old granddaughter is being taught how to craft Byssus from raw materials, but training takes decades.[33] Chiara has repeatedly expressed her complete rejection of writing down certain parts of the process and profiting off the skill. "...tried to force me to charge entrance fees and write down my patterns and secrets. But I will defend this sacred oath with my fingernails as long as I breathe!"[34] Does the Italian coastguard have a moral right to the knowledge? What happens if the Pinna nobilis clam makes a comeback and has a thriving population? At what point does the cultural ownership of information end?

6. Conclusion

Our proposed approach offers a promising path to preserving cultural heritage through ethical AI-human collaboration. By judicious use of archival footage, abiding by strict principles that govern data, and honoring the rights and wishes of traditional craftspeople, we can train AI systems to document endangered practices and empower communities to maintain control over their cultural knowledge. This approach has the potential to enrich our collective human experience, ensuring that diverse cultural practices and traditional craftsmanship continue to thrive in the age of AI.

References

[1] Carroll, S.R., Herczog, E., Hudson, M. et al. Operationalizing the CARE and FAIR Principles for Indigenous data futures. Sci Data 8, 108 (2021). https://doi.org/10.1038/s41597-021-00892-0

[2] Cooperman, J. Turning Spit to Silk. In: The Common Reader: A Journal of the Essay; 2023 April 17 https://commonreader.wustl.edu/turning-spit-to-silk/ Accessed 2025 January 31.

[3] Hudson M, Carroll SR, Anderson J, Blackwater D, Cordova-Marks FM, Cummins J, David-Chavez D, Fernandez A, Garba I, Hiraldo D, Jäger MB, Jennings LL, Martinez A, Sterling R, Walker JD and Rowe RK (2023) Indigenous Peoples' Rights in Data: a contribution toward Indigenous Research Sovereignty. Front. Res. Metr.Anal., 03 May 2023 Sec. Scholarly Communication. Volume 8-2023| https://doi.org/10.3389/frma.2023.1173805

[4] Counts, A. YouTubers Are Selling Their Unused Video Footage to AI Companies, In: Bloomberg, 2025 Jan 10 https://www.bloomberg.com/news/articles/2025-01-10/youtubers-are-selling-their-unused-video-footage-to-ai-companies Accessed January 31, 2025.

[5] Weatherbed, J. Adobe Premiere Pro now lets you find video clips by describing them: 'Media intelligence' and caption translations are launching alongside new playback features for After Effects. In: The Verge. 2025 Jan 22. https://www.theverge.com/2025/1/22/24349299/adobe-premiere-pro-after-effects-media-intelligence-search Accessed January 31, 2025.

[6] Dorey, F. Homo sapiens – modern humans. In: Australian Museum. 2020 October 16. https://australian.museum/learn/science/human-evolution/homo-sapiens-modern-humans/ Accessed January 31, 2025.

[7] Stein, E. The last bridge master. In: BBC. 2024 Dec 10. https://www.bbc.com/travel/article/20241206-the-last-inca-bridge-master Accessed January 31, 2025.

[8] Parchment and Vellum Making. In: Heritage Crafts. 2024 https://www.heritagecrafts.org.uk/craft/parchment-and-vellum-making/ Accessed January 31, 2025.

[9] Severn, T. The Brendan Voyage: Sailing to America in a Leather Boat to Prove the Legend of the Irish Sailor Saints. 2000 April 4. Random House Publishing Group.https://www.amazon.com/Brendan-Voyage-Sailing-America-Exploration/dp/0375755241 Accessed January 31, 2025.

[10] 1 Kakutai, T., Campbell- Kamariera, K.,Mead, A., Mikaere, K., Moses, C., Whitehead, J., & Cormack, D. (2023) Māori data governance model. Te Kāhui Raraunga. https://www.kahuiraraunga.io/_files/ugd/b8e45c_803c03ffe532414183afcd8b9ced10dc.pdf p. xi.

[11] Redden J, Brand J, Terzieva V. Data Harm Record (Updated) [Internet]. Updated August 2020. Available from: https://datajusticelab.org/data-harm-record/

[12] Victorian Farm (TV Mini Series 2009), https://www.imdb.com/title/tt1372211/

[13] Miangoleh, S.M. H, Reddy, M., Aksoy, Y. Scale-Invariant Monocular Depth Estimation via SSI Depth. In: AIModels.fyi. 2024 Jun 13. https://www.aimodels.fyi/papers/arxiv/scale-invariant-monocular-depth-estimation-via-ssi Accessed January 31, 2025.

[14] Miangoleh, S. M.H., Reddy, M., Aksoy, Y. Scale-Invariant Monocular Depth Estimation via SSI Depth. In: Simon Fraser University. Computational Photography Lab. https://yaksoy.github.io/sidepth/ Accessed January 31, 2025.

[15] Vision-guided Robotics – How Cameras are Transforming Robotics. In: TechNexion. https://www.technexion.com/resources/vision-guided-robotics-how-cameras-are-transforming-robotics/ Accessed January 31, 2025.

[16] Griffin, B. Computer Vision Meetup: Learning Robot Perception and Control using Vision with Action. 2024 May 8. https://www.youtube.com/watch?v=qVRRHSZk5jM Accessed January 31, 2025.

[17] Seita, D., Mahler, J., Danielczuk, M., Matl, M. Drilling Down on Depth Sensing and Deep Learning. In: Berkeley Artificial Intelligence Research. 2018 Oct 23. https://bair.berkeley.edu/blog/2018/10/23/depth-sensing/ Accessed Janaury 31, 2025.
[18] Zhao, T., Tompson, J., Driess, D., Florence, P., Ghasemipour, K., Finn, C., Wahid, A. ALOHA Unleashed: A Simple Recipe for Robot Dexterity In: Google DeepMind 2024 Sep 12. https://deepmind.google/discover/blog/advances-in-robot-dexterity/ Accessed January 31, 2025.
[19] Hudson, M., Carroll, S.R., Anderson, J., Blackwater, D., Cordova-Marks, F., Cummins, J., David-Chavez, D., Fernandez, A., Garba, I., Hiraldo, D., Jäger, M.B., Jennings, L.L., Martinez, A., Sterling, R., Walker, J.D.,Rowe, R.K. Indigenous Peoples' Rights in Data: a contribution toward Indigenous Research Sovereignty. In: Frontiers in Research Metrics and Analytics. Volume 8 - 2023 | https://doi.org/10.3389/frma.2023.1173805 Accessed January 31, 2025.
[20] Catching Fish With Only A Spider Web & Kite 2021 https://www.youtube.com/watch?v=FY4f6Uc9zK8 Accessed January 31, 2025.
[21] Aura Humanoid to Interact With Sphere Guests in Las Vegas. In: Robotics247.com 2023. Sep 19. https://www.robotics247.com/article/aura_humanoid_interact_sphere_guests_las_vegas Accessed January 31, 2025.
[22] https://maineshakers.com/about/ Accessed January 31, 2025.
[23] Why are there only two Shakers left? In: The Times of India 2024 Sept 10. https://timesofindia.indiatimes.com/world/us/why-are-there-only-two-shakers-left/articleshow/113230307.cms Accessed January 31, 2025.
[24] In: Health and Safety Executive. https://www.hse.gov.uk/lung-disease/stonemasonry.htm Accessed January 31, 2025.
[25] Kim, B., Yoon, J., Choi, B., Shin, Y.C. In: National Library of Medicine. 2013. Elsevier B.V. https://pmc.ncbi.nlm.nih.gov/articles/PMC3889078/, organic solvents, https://pubmed.ncbi.nlm.nih.gov/2245185/) Accessed January 31, 2025.
[26] In: Center for Disease Control 2024, Mar 22. https://www.cdc.gov/arthritis/basics/index.html Accessed January 31, 2025. Accessed January 31, 2025.
[27] Schilling, M. Quirky: The Remarkable Story of the Traits, Foibles, and Genius of Breakthrough Innovators Who Changed the World. 2019 Sep 10. Public Affairs.
[28] All Hallucinations Are Not Bad: Acknowledging Gen AI's Constraints and Benefits: NTT DATA point of view on Generative AI, examining the concept of hallucination, its implications, and responsible AI practices.In: NTT DATA. 2024. https://www.nttdata.com/global/en/insights/focus/2024/not-all-hallucinations-are-bad-the-constraints-and-benefits-of-generative-ai Accessed January 31, 2025.
[29] Fleisher, C. Kill the machines: To what extent can technological disruption lead to civil unrest? In: American Economic Association. 2020, Sep 8. https://www.aeaweb.org/research/thresher-machine-england-civil-revolt Accessed January 31, 2025.
[30] In: X. 2024 Sep 15. https://x.com/Rahll/status/1835317659082133833 Accessed January 31, 2025
[31] Paradiso, M. The Last Woman Who Makes Sea Silk. In: BBC. 2015, Sep 2. https://www.bbc.com/news/magazine-33691781 Accessed January 31, 2025.
[32] Chef Evan Funke Makes Trofie al Pesto Genovese with Jimmy Kimmel & Guillermo In: Jimmy Kimmel Live. 2024 https://www.youtube.com/watch?v=QSGj6nVZxFU Accessed January 31, 2025.
[33] Cavalletti, I. Chiara Vigo: the Master of Byssus Silk. In: eco-nnect.com 2024, May 24. https://eco-nnect.com/chiara-vigo-byssus-silk/ Accessed January 31, 2025
[34] Stein, E. The last surviving sea silk seamstress. In: BBC. 2017, Sep 6 https://www.bbc.com/travel/article/20170906-the-last-surviving-sea-silk-seamstress?zephr-modal-register Accessed January 31, 2025.